Exp. No: 13

PERFORMANCE EVALUATION OF DENSE WDM MULTIPLEXING AND DEMULTIPLEXING USING OPTISYSTEM

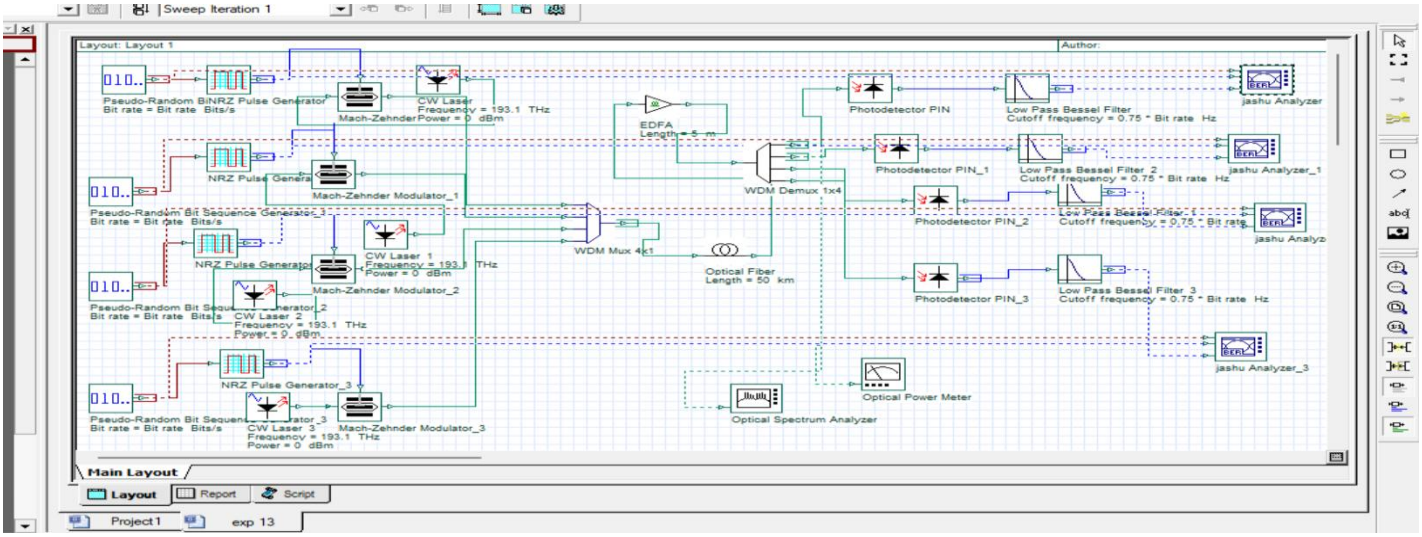
OBJECTIVES

1. To design and simulate a Dense Wavelength Division Multiplexing (DWDM) system in OptiSystem and analyze its performance under different test conditions.
2. To analyze the effect of wavelength (channel) spacing, fiber length (loop count) and channel power on BER, Q-factor, OSNR and crosstalk.
3. To compare modulation formats (NRZ, RZ, QPSK) and determine the optimal optical amplifier (EDFA) placement for long-haul DWDM transmission.
4.

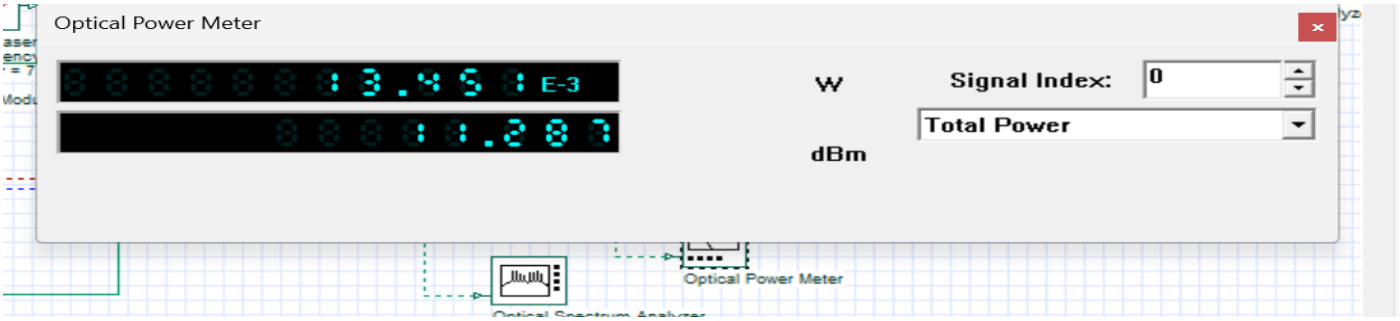
SOFTWARE REQUIRED

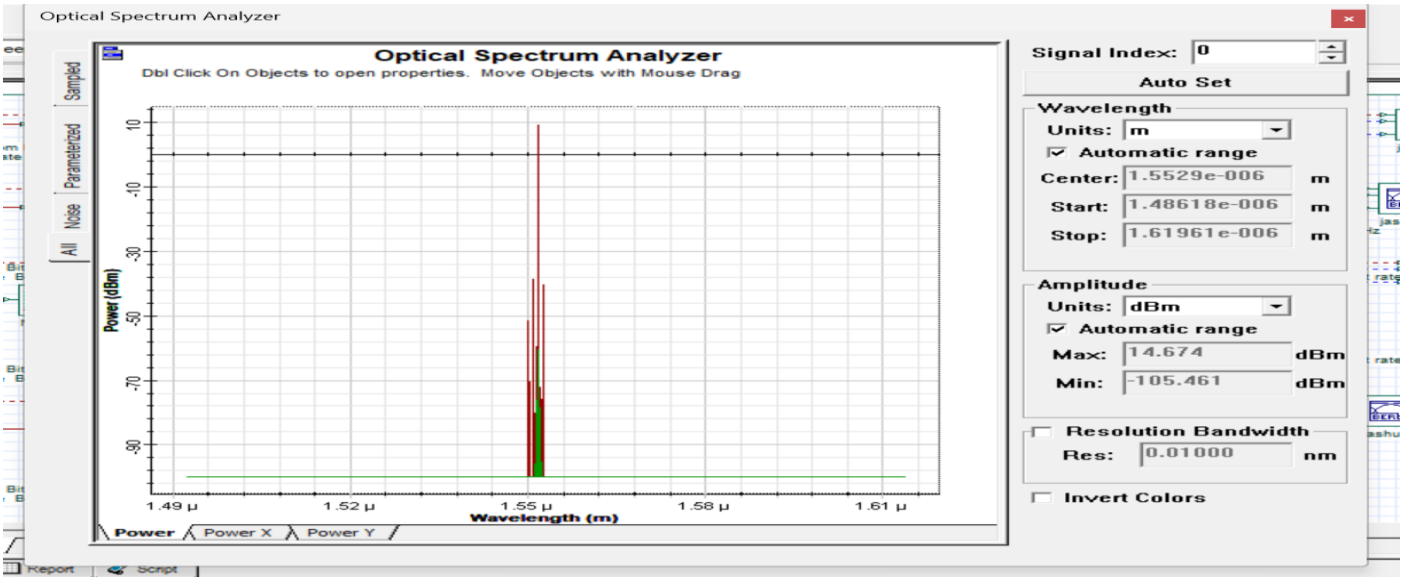
- OptiSystem

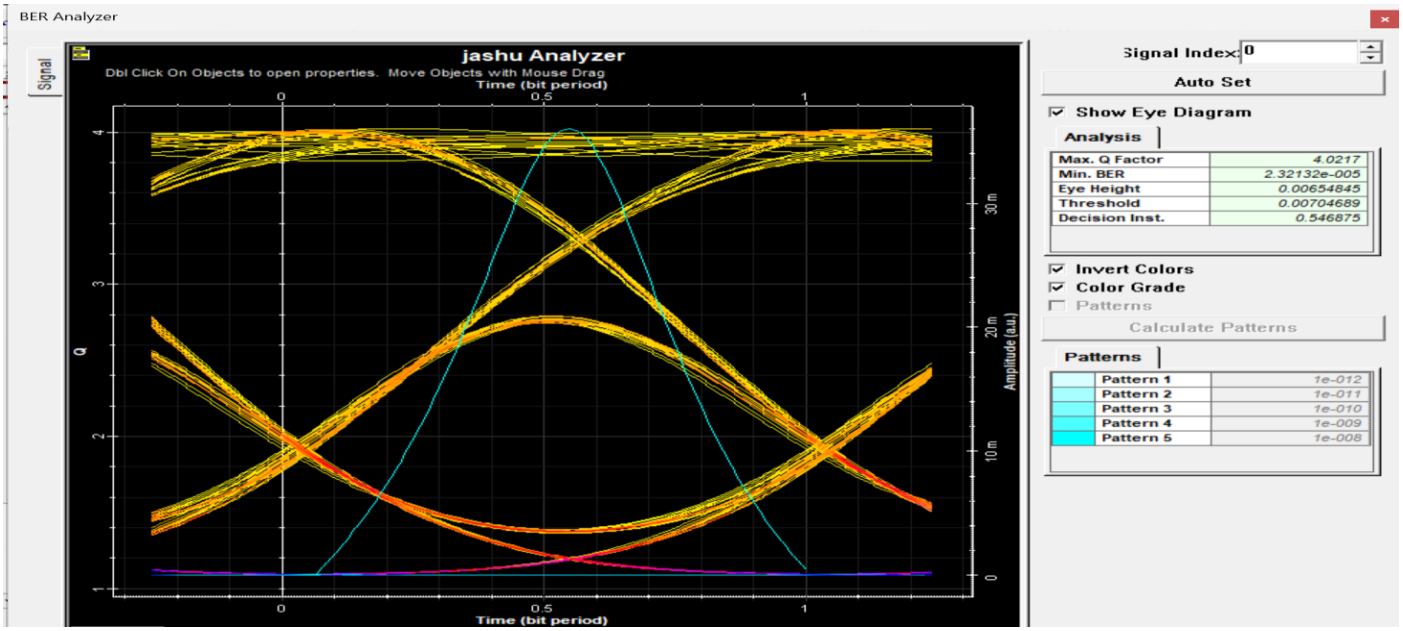
BLOCK DIAGRAM

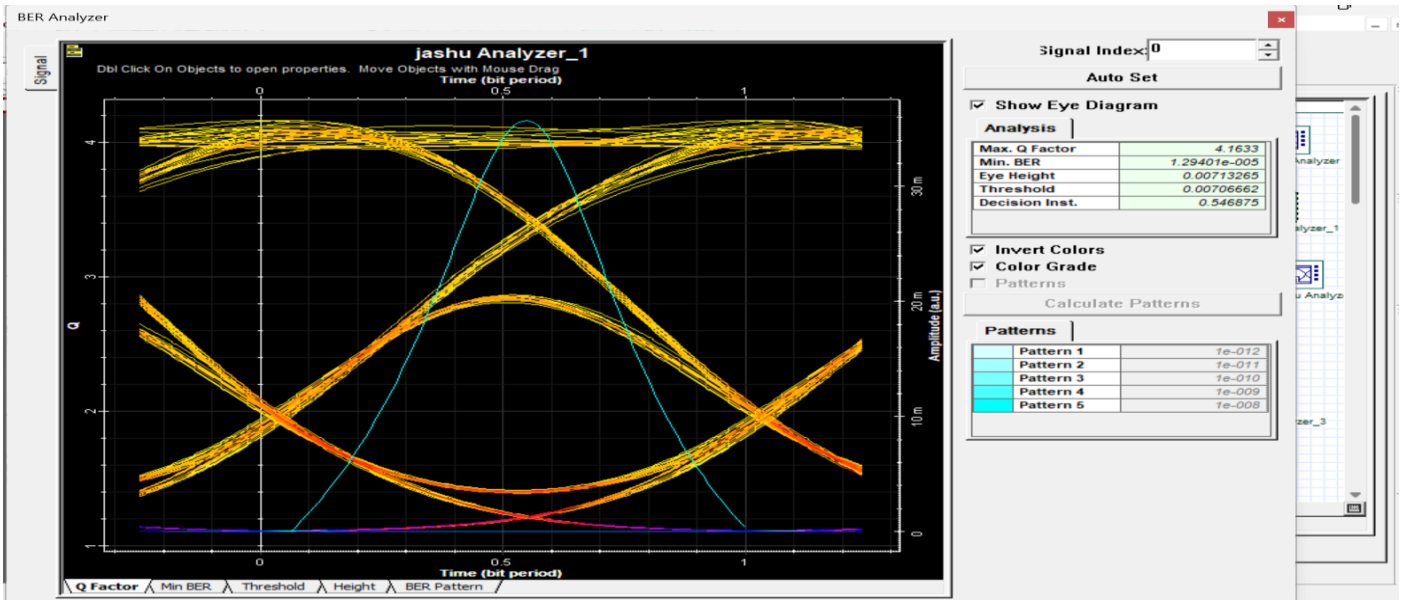


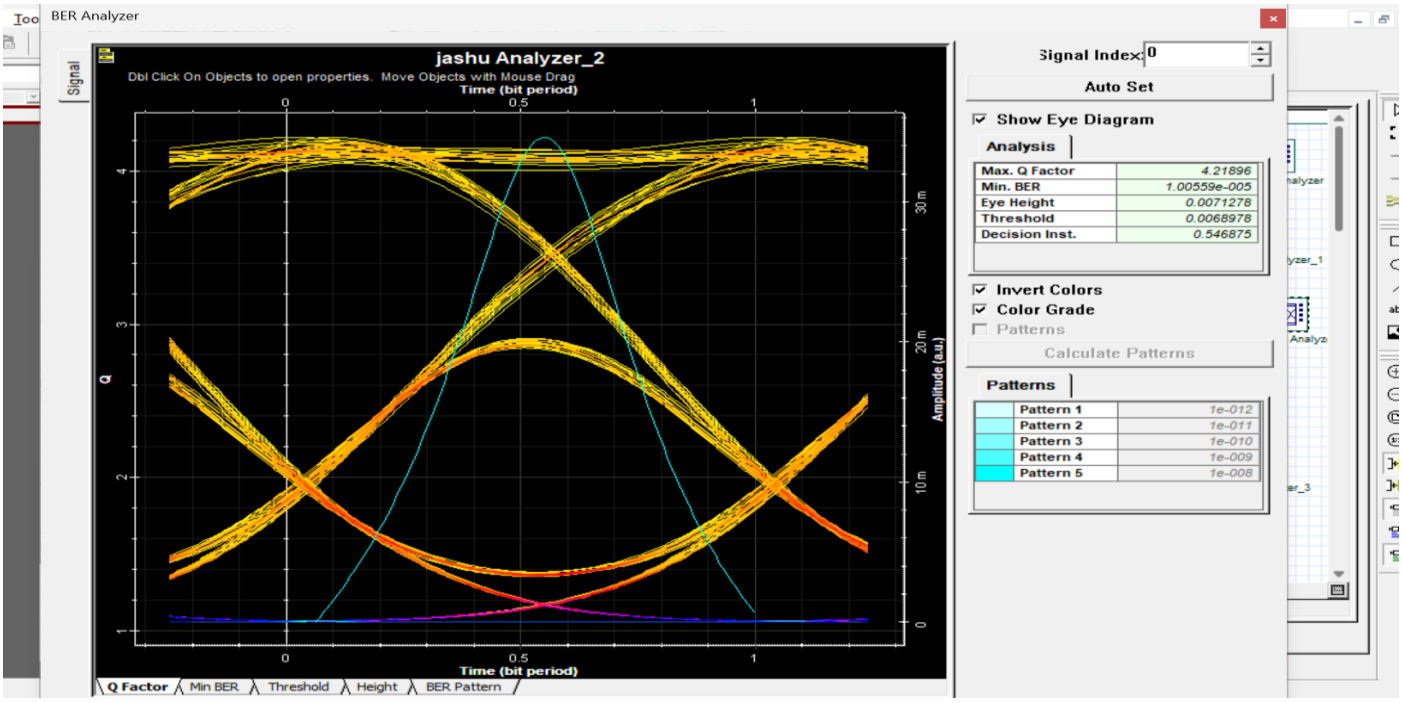
SAMPLE OUTPUT WAVEFORMS

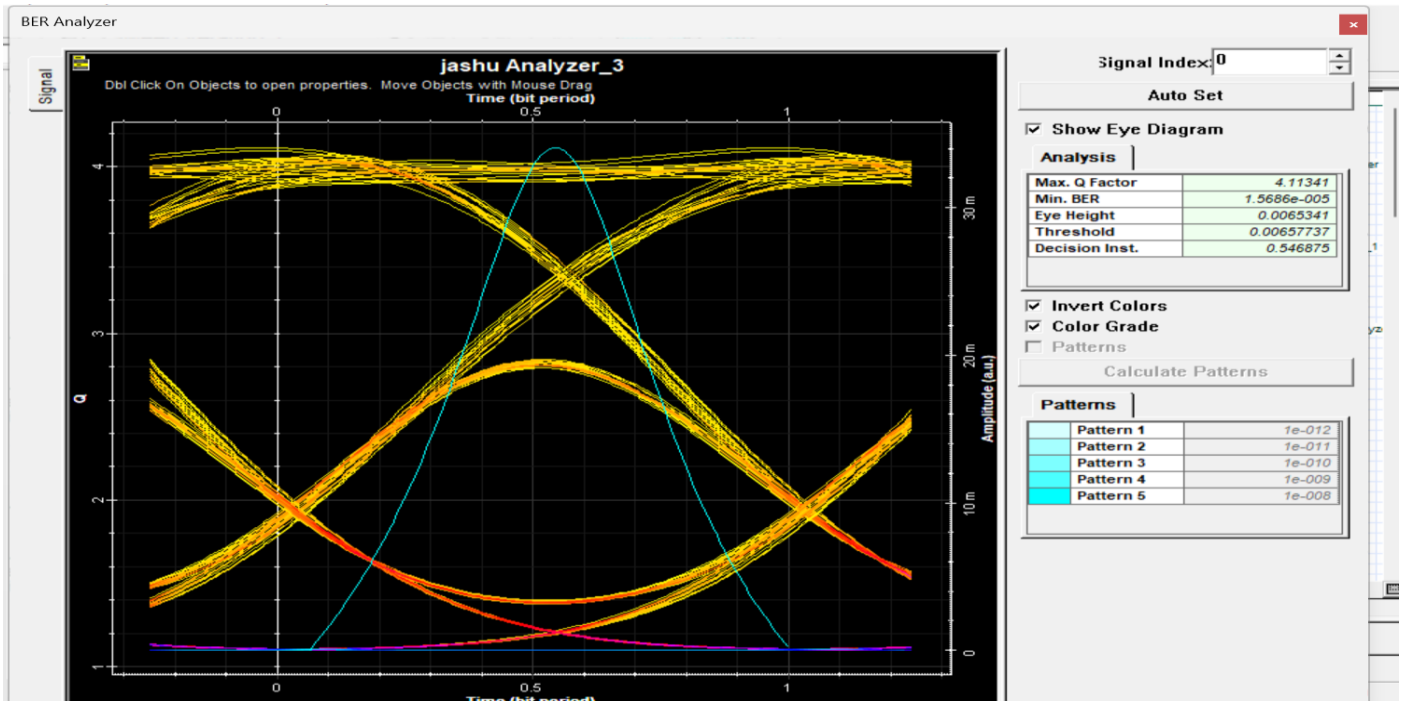





BER Analyzer
jashu Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Q
Amplitude (a.u.)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.0217
Min. BER 2.32132e-005
Eye Height 0.00654845
Threshold 0.00704689
Decision Inst. 0.546875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008


BER Analyzer
jashu Analyzer_1
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Q
Amplitude (a.u.)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.1633
Min. BER 1.29401e-005
Eye Height 0.00713265
Threshold 0.00706662
Decision Inst. 0.546875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


BER Analyzer
jashu Analyzer_2
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Q
Amplitude (a.u.)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.21896
Min. BER 1.00559e-005
Eye Height 0.0071278
Threshold 0.0068978
Decision Inst. 0.546875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern



Reflection: I learned how multiple optical signals are transmitted through a single fiber using different wavelengths. I understood the importance of WDM in increasing communication capacity.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & DWDM Setup	Correct selection and connection of OptiSystem components for the DWDM link (CW Laser Array, Modulator/Pulse Generator, WDM MUX, Optical Fiber with Loop Control, EDFA/Raman Amplifier, WDM DEMUX, PIN/APD Photodetectors, BER Analyzer, Optical Spectrum Analyzer, Optical Power Meter).	All transmitter, MUX, fiber loop, amplifier, DEMUX and receiver blocks are correctly selected, connected in sequence and configured without errors.	All components are present with minor connection or layout errors, but the DWDM link remains functional.	Component selection is partially complete; missing analyzers or amplifier/loop errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Test Case Execution	Accurate setup of ITU-grid frequencies (193.1 THz onwards), channel spacing (100/50/25 GHz), loop count (1/5/10 loops of 50 km), modulation (NRZ/RZ/QPSK), EDFA spacing (50/100/150 km), channel power (0–10 dBm), fiber attenuation	All parameters are accurately set and all five test cases are executed completely.	Parameters are mostly accurate with minor setting discrepancies; one test case is incomplete.	Basic parameters are configured, but multiple test cases or parameter values are missing.	Incorrect parameter settings or failure to run the required test cases.	20%

	(0.2 dB/km), dispersion (16 ps/nm/km) and amplifier noise figure (5 dB).					
Measurement & Performance Monitoring	Accurate extraction and recording of BER, Q-factor, OSNR, crosstalk and power penalty for every test case, using the BER Analyzer, Optical Spectrum Analyzer and Optical Power Meter.	Precise data extraction across all test cases, neatly tabulated with complete BER, Q-factor and OSNR values and clear spectra and eye diagrams.	Data is collected with minor omissions in BER, Q-factor, OSNR or crosstalk readings.	Partial data collection with incomplete tables or missing spectrum/eye diagram captures.	Inaccurate, incomplete or missing data and measurements.	20%
Performance Analysis	Detailed evaluation of how spacing, distance, modulation format, amplifier placement and channel power affect crosstalk, dispersion, ASE noise, SPM/XPM nonlinearity and spectral efficiency.	Comprehensive analysis correctly relating each parameter to BER, Q-factor and OSNR with sound physical explanations and trade-offs.	Clear analysis identifying basic trends in BER, Q-factor and OSNR for most test cases.	Basic attempt at result analysis without explaining physical mechanisms such as crosstalk, dispersion or nonlinear effects.	Erroneous or missing analysis of simulation results.	20%
Report & Conclusion	Identification of optimal spacing, amplifier interval, modulation format and power level; structure and clarity of the report.	Report features clear spectra and eye diagrams, complete result tables and insightful conclusions identifying optimal DWDM design parameters.	Satisfactory report with relevant displays and a sound concluding summary.	Incomplete report with missing graphs, readings or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

Result: Dense WDM multiplexing and demultiplexing were simulated and verified using OptiSystem.